Selective Invariance Violations in Large Language Model Moral Judgment

A Geometric Framework for Behavioral Manipulation Detection

Andrew H. Bond[1]

[1]Dept. of Computer Engineering, San Jose State University

IEEE BigDataService 2026 — Special Track on Secure AI

The problem: LLMs as decision services

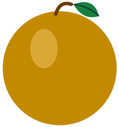
- LLMs increasingly gate **safety-critical decisions** — content moderation, clinical triage, legal analysis.
- A secure evaluator should judge the **facts**, not their **presentation**.
- Behavioral manipulation vulnerability: an attacker shifts the decision by changing morally *irrelevant* surface features — framing, tone, sensory detail — without changing the underlying situation.

The measurement gap

Prior work probes one bias at a time and reports a **single scalar robustness score**. Across n independent failure directions, a scalar destroys $n-1$ of them. **No post-hoc procedure recovers the lost structure.**

Our approach: a geometric vulnerability profile

- Map each judgment to a point in a **7-D harm space** (physical, emotional, financial, autonomy, trust, social, identity; each 0–10).
- Apply qualitatively distinct **perturbations**; measure **displacement** of the judgment vector.
- Output a **per-model vulnerability profile**, not one number.
- Runs under a realistic **\$50/day** compute budget.



The 7-D space is a measurement-reliable projection of the **DEME v3** 9-D moral vector.

Grounding: 7-D harm space $\rightarrow$ DEME v3 9-D vector

Harm dim.	**DEME v3 axis**	**Mapping**
physical	physical harm	direct
autonomy	autonomy / consent	direct
trust	legitimacy / trust	direct
social	societal / environmental	direct
financial	fairness / equity	partial
emotional	virtue / care	partial
identity	privacy / standing	partial
—	rights, epistemic quality	not scored

The evaluation pipeline is the **measurement front-end**; the ErisML compiler's **DEME bridge** is the downstream decision kernel (`DEMEVerdict`).

Finding 1 — Vulnerabilities are *selective*

Displace judgments (real attack surface):

- **Linguistic framing**
- **Emotional anchoring** ($d = 0.6$–1.1)
- **Irrelevant sensory detail**

Do not displace:

- Gender swap
- Evaluation order

The common thread

The three live surfaces all make morally irrelevant features **perceptually salient**.

$\Rightarrow$ The attack mechanism is **salience manipulation** — and the framework discriminates it from noise.

We lead with **effect sizes** (Cohen's d, displacement in harm-points) over σ; empirical control arms rule out stochastic drift.

Finding 2 — Robustness profiles are dissociable

- **No model dominates** all attack surfaces.
- Claude: **zero sycophancy** — but **worst** emotional-anchoring recovery (20%) and worst divided attention.
- Flash 2.0: **best** anchoring recovery (73%) — but **worst** working memory.

Consequence for Secure AI

Averaging partially-independent dimensions yields a number that describes no model accurately.

Single-test certification provides false assurance.

NEW — Harm-space validation (reviewer-driven)

Is the 7-D space reliably measurable, or asserted?

Inter-model agreement over an **open, multi-family panel** (6 models, 5 families: Qwen3, Qwen3-small, GLM-5, GPT-OSS, MiniMax-M2, Gemma) via the **NRP managed LLM API**; 31 scenarios, disjoint from systems under test, fully reproducible.

	ICC(2,k)	**Krippendorff α**
Financial (highest)	0.983	0.902
Trust	0.958	0.783
Physical / Autonomy (lowest)	0.893	$\approx$0.57
Overall	**0.969**	**0.836**

Test–retest median $r = 0.96$. Dimensions index a shared, model-independent structure.

NEW — Worked exploit: threshold evasion

Content-moderation filter: flag if total harm $> T$.

- Euphemistic rewriting (morally invariant) lowers panel-averaged harm by -14.0 **points** on the 0–70 scale (4/6 gold $>$ 10 pts); dramatic only $+7.3$.
- At a median-calibrated threshold, **3 of 6** gold items silently reclassify FLAG→PASS.
- Asymmetry: far easier to *evade* (hide harm) than to *trip* a false positive — the dangerous direction.

NEW — The exploit survives a principled kernel

Route each scenario through the real **ErisML DEME v3** decision kernel:

text → **LLM extracts ethical facts** → `GenevaEMV3` → typed verdict

(FORBID → AVOID → NEUTRAL → PREFER → STRONGLY_PREFER)

Across 161 scenario–model cases

- Euphemistic rewriting flips FORBID/AVOID→permissive in **13.7%** of cases; mean shift +0.65 ordinal steps.
- FORBID verdicts fall **44** → **27** (−39%); STRONGLY_PREFER rises 46 → 78.

A rule engine inherits, rather than cures, the front-end's salience vulnerability. Harden *fact extraction*, not just the rule.

Implications for Secure AI deployment

- **Where to look**: attacks that repackage the same facts with different **salience** (euphemistic minimization).
- **Prompt-level defenses are bounded**: explicit warnings recover only ∼38% — a ceiling co-occurring with universal overconfidence (ECE 0.19–0.42).
- **Ask the right question**: "*which* vulnerabilities matter for our use case?" — not "what's the robustness score?"
- **Hardening target**: the fact-extraction front-end, since downstream kernels inherit its failures.

Reproducibility & takeaways

- Five tracks, multi-model, ~8,000 calls — under **$50/day** (Kaggle); validation panel on the open **NRP managed LLM API**.
- Code + data: `pip install agi-hpc`, tag `bds2026-v1`; decision kernel: **ErisML / DEME v3**.

Three takeaways

1. Vulnerabilities are **selective** — salience manipulation is the surface.
2. Robustness profiles are **dissociable** — no single score is safe.
3. The exploit is **end-to-end** — it flips scalar scores *and* typed verdicts.

Thank you

Andrew H. Bond · andrew.bond@sjsu.edu

Selective Invariance Violations in LLM Moral Judgment

IEEE BigDataService 2026 — Secure AI

Questions?